

Exploring the Correlation between Cross-Lingual Alignment and the Generative Capabilities of Large Language Models

1. Abstract

Large language models (LLMs) are often trained on large amounts of English data, but they can still understand and generate text in many other languages. In this project, we will investigate whether the amount of training data available for a language affects how well an LLM performs in that language. We will also examine whether differences in how languages are represented in models can help explain performance differences across languages.

First, we will reproduce an existing experiment that measures how closely different languages are represented within different layers of a transformer model. We will examine whether languages with more similar representations are more likely to transfer information to one another successfully.

Second, we will reproduce an activation patching experiment in which we replace parts of a model's representation of a non-English input with the corresponding English representation. We will then observe whether this changes the model's output. If time permits, we will extend our investigation by testing models on tasks such as mathematical reasoning, open-domain question answering, summarization, and translation. We will compare performance across high-resource and low-resource languages to identify differences in accuracy and consistency. Overall, this project aims to understand better how LLMs represent different languages and how these representations may affect multilingual performance.

3. Research Background

Multilingual large language models are often trained primarily on English data but remain competent in other languages, completing NLU (Native Language Understanding) tasks in languages they are less familiar with (Touvron et al., 2023). However, lower alignment may increase the risk that a model will solve problems ineffectively.

Ravisankar et al. (2026, which we are reproducing, sought to answer two questions: Does the cross-lingual alignment of a model correlate with the accuracy with which it completes natural language understanding problems? Moreover, does misalignment in English cause transfer failure in other languages? It found that transfer successes were more closely aligned with English text, and that when a patch was semantically equivalent in English and the target language, it was more likely to solve problems accurately in that language. It also found that languages more closely related to English had a stronger correlation (Ravisankar et al., 2026). We are looking to replicate this result and demonstrate that NLU predictions correlate with representation alignment across languages (i.e., whether a model encodes information in a way similar to humans).

A limitation of the study is that it can only test the hypothesis via discriminative NLU tasks (predicting based on an existing input, not writing from scratch), because translating to a language requires an English input, and vice versa. Additionally, since human translators were used to create the original translations against which the models were compared, there is a possibility of artifacts. It also did not test correlations between non-English languages, as all tests were conducted between English and another language, and only a few languages were used. Further comparisons, including those involving low-resource languages or non-English languages, would help expand on their findings.

Improving the effectiveness of models in languages other than English is incredibly valuable to anyone who might use an LLM without speaking English, or to those who wish to learn a new language via an LLM. As such, understanding the correlation between a model's effectiveness in one language and its effectiveness in others can provide a reliable, useful way to measure its effectiveness in certain languages without directly testing it in those languages.

4. Methodology

The methodology will consist of 3 stages: (1) literature review, (2) reproduction of the original experiments on a smaller scale, and (3) extension to the generative task.

1. Data

For the literature review stage, we will start with papers suggested by the study's project leader. Then we will conduct further research using keywords related to the topic via Google Scholar and the ACL Anthology.

For the reproduction stage, we will use the multilingual benchmark datasets used in the original paper: Belebele (Bandarkar et al., 2024), Xstudycloze (Lin et al., 2022), and Xscopa (Ponti et al., 2020).

1. Belebele for reading comprehension (“Can you understand what you read?”)
 - Dataset link:
https://github.com/facebookresearch/belebele?utm_source=chatgpt.com
 - Paper: Bandarkar et al. (2024)
 - Task: Multiple-choice reading comprehension
 - Languages: 122 language variants
 - Size: 109,800 questions total — 900 per language variant
 - Source: passages from FLORES-200.
2. XStoryCloze for narrative and commonsense reasoning (“Can you understand what should happen next in a story?”)
 - Dataset Link:
https://github.com/facebookresearch/fairseq/blob/main/examples/xglm/XStoryCloze.md?utm_source=chatgpt.com
 - Paper: Lin et al. (2022)
 - Task: Story completion / commonsense reasoning
 - Languages: English + 10 additional languages
 - Original dataset: StoryCloze
 - Purpose: Evaluate multilingual zero-shot and few-shot capabilities.
3. XCOPA for causal commonsense reasoning (“Can you understand why something happened or what it causes?”)

- Dataset link: https://github.com/cambridgeltl/xcopa?utm_source=chatgpt.com
- Paper: Ponti et al. (2020)
- Full name: XCOPA: A Multilingual Dataset for Causal Commonsense Reasoning
- Task: Choose the most plausible cause or effect
- Languages: 11 languages.
- XCOPA is based on COPA, the Choice of Plausible Alternatives task.

For the optional extension, we will investigate datasets appropriate for generative tasks, including mathematical reasoning, open-domain question answering, summarization, and Translation. The specific datasets will be selected based on their multilingual coverage, suitability for evaluating generation, and the availability of computational resources.

2. Methodology

First, we will conduct a literature review of approximately 20 relevant research papers covering multilingual LLMs, low-resource languages, cross-lingual alignment, and representation realignment. We will follow the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework to systematically identify, screen, and select relevant studies ([PRISMA, n.d.](#)). The literature review will be used to understand existing approaches and identify the research gap that the project addresses.

Second, we will reproduce the code repository given in the paper “[Can you map it to English? The Role of Cross-Lingual Alignment in the Multilingual Performance of LLMs](#)” (Ravisankar et al., 2026) with a smaller datasets and LLM model. All the tries and technical barriers we encountered, with solutions (including those that remain unresolved), will be noted in detail.

Lastly, we will try to contribute to something new by extending the evaluation to generative tasks such as Mathematical reasoning, Open-domain QA, summarization, and Translation by modifying the existing code and analyzing the resulting performance

3. Experiment

The reproduction section will consist of 2 main experiments.

Experiment 1: Cross-Lingual Alignment

We will compare Cross-Lingual alignment between TS and TF across different transformer layers. This will allow us to examine whether stronger alignment is associated with successful multilingual transfer.

Experiment 2: Activation patching

We will also reproduce the activation patching experiment by replacing selected non-English activations with their corresponding English counterparts and examining whether this intervention changes the model's sections. An unrelated English example will be used as a control condition. If the reproduction is completed before the final deadline of the project, the third experiment will extend the evaluation to multilingual generative tasks such as: Mathematical reasoning, Open-domain QA, Summarization, and Translation.

4. Evaluation

Based on the paper “[Can you map it to English? The Role of Cross-Lingual Alignment in the Multilingual Performance of LLMs](#)” (Ravisankar et al., 2026), the primary evaluation metrics will include:

1. Task accuracy on 3 different multilingual benchmark datasets: Belebele (Bandarkar et al., 2024), Xstorycloze (Lin et al., 2022), Xcopa (Ponti et al., 2020)
2. DALI, DALIst, and MEXAT for measuring cross-lingual representation alignment.
3. Layer-wise alignment to examine alignment across transformer layers.
4. Transfer Success and Transfer Failure rates to measure multilingual transfer performance.
5. Activation-patching flip rate to measure how often an incorrect prediction becomes correct after patching.

Further generative evaluation will be considered based on the selected tasks and dataset.

5. Comparison

The primary baseline will be the experimental methodology and findings reported in the original paper. Our reproduced results will be compared with the published results to assess the consistency and reproducibility of the findings, as well as the original settings and resource constraints in our reproduction.

If the generative evaluation extension is completed, we will compare the results with the reproduced baseline to examine whether the observed relationship between cross-lingual alignment and

multilingual performance extends to generative tasks, including multilingual summarization, mathematical reasoning, and open-domain question answering.

5. Significance

Most LLMs are trained mostly in English, with a few limited ones in other languages, but people use them in hundreds of different languages. It is not yet fully understood how they manage to achieve this or why the approach works. The current explanation shows the model mapping non-English input to an internal English representation and solving the problem in this way. If this is the right method, then a model's ties in a specific language depend on how well that rendering back to English holds up. A failure in other languages would mean a misalignment in their internal representations. Ravisankar et al. (2026) provided evidence for this with 8B-parameter models using multiple-choice benchmarks, in which the model is given a passage and a set of answer options and must select the correct one. For those whose primary language is not English, this would determine the reliability of models, which can assist in either understanding English as a required parameter or another method.

We reproduce the original findings using limited computational settings, smaller models, and reduced data. The original study ran on 8B-parameter models, and the computational cost meant they only performed the causal analysis on one model. Testing the same relationship on models running on standard hardware will show whether the results hold. If it does not hold, the findings will shift the claim onto specifically larger models. Translation is a generative task and a different approach from the original multiple-choice format, with no single correct output. The evaluation will be more difficult, reflecting how the models are actually used, and the testing phase will determine whether the relationship is more concerned with the produced text or with a selected option.

For most languages, there simply is no benchmark written in that language, due to the massive expense involved. Across the vast language spectrum of over 7,000 languages, only a minimal number are currently represented in these technologies (Joshi et al., 2020), and those not represented in these technologies have the least training data, leading to the worst performance in models.

Cross-lingual alignment only needs parallel text and access to the model's internals, so if it reliably tracks accuracy, it could estimate how a model performs in a language without even building a benchmark. This is most beneficial to users of languages without a measure in place and to developers choosing where to spend limited resources.

These results also influence the work on realignment, which attempts to reduce the distance between a model's and non-English representations (Liu & Niehues, 2025). That work needs to know where in the model to intervene, and measuring alignment layer by layer shows where the gap opens up. Ravisankar et al. (2026) also discuss guiding the model's representations during use rather than retraining it, but this creates risks. The model may become more confident, but still give the wrong answer. Always pushing towards English could erase another language's edge. More work could address the limits, specifically that alignment is measured only between English and another language, with none between two non-English languages.

6. SMART Timeline

1. Literature Review - 2 weeks

Making a document summarizing the current research in the area. Make a weekly meeting to discuss our progress, address questions, and talk about the papers we reviewed during the previous week with the project leader.

2. Baseline Reproduction - 4–8 weeks

Clone the repository and set up the required environment using smaller models and dataset subsets based on computational availability. Then the evaluation process.

3. Evaluation Scope Extension - 4 weeks (Optional)

Modify the existing code to support these evaluations and analyze the resulting performance in generative tasks (such as Mathematical reasoning, Open-domain QA, Summarization, Translation).

7. References

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